Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Entropy is not continuous in standard probability spaces

Files.

- File: `briefs/EntropyTopology/EntropyTopology.lean`
- Library: `EntropyTopology`
- Imports: `Mathlib.Analysis.SpecialFunctions.BinaryEntropy`, `Mathlib.Analysis.SpecialFunctions.Log.Base`, `ForMathlib.Analysis.SpecialFunctions.Log.NegMulLog`, `ForMathlib.MeasureTheory.Integral.DominatedConvergence`, `ForMathlib.MeasureTheory.Measure.DisjointSeq`, `ForMathlib.MeasureTheory.Measure.TotalVariation.Basic`

Setting. A measure space (X, μ) (`MeasureTheory.Measure`). The predicate `IsDensity` says that $\rho\colon X \to \mathbb{R}$ is measurable, pointwise nonnegative, integrable with $\int \rho\, d\mu = 1$, and that $\rho \log \rho$ is integrable. This is the space P_μ; as a type it is `Density`. The entropy S and the Jensen–Shannon divergence are in bits, and $d_1(\rho, \sigma) = \int |\rho - \sigma|\, d\mu$. The probability measure p_ρ with density ρ is `prob` $\mu\, \rho$, and the total variation distance is Mathlib's `tvDist`, i.e. $|p_\rho - p_\sigma|(X)$.

For a divergence D there are two topologies "generated by the balls" $\{\sigma : D(\sigma,\rho) < \delta\}$. In `divTopology` D a set is open iff it contains a ball around each of its points; `ballTopology` D is generated by declaring all balls open. For a metric they agree; in general the second is finer. With D = JSD they are `jsdTopology` and `jsdBallTopology`; `tvTopology` is `divTopology` of $\mathrm{tvDist}(p_\sigma, p_\rho)$. The $L^1(\mu)$ topology `l1Topology` is induced by $\rho \mapsto [\rho] \in L^1(\mu)$ (`Density.toL1`).

Two results go beyond the brief and answer the questions it leaves open. A measure *has a disjoint sequence* (`Measure.HasDisjointSeq`, ForMathlib) if there are pairwise disjoint measurable sets A_n, $n \in \mathbb{N}$, with $0 < \mu(A_n) < \infty$. A topology on P_μ is *admissible* (`Admissible`) if mixing $[0,1] \times P_\mu \times P_\mu \to P_\mu$ (with the product topology) and S are continuous. The *entropy topology* `entropyTopology` is induced by $\rho \mapsto ([\rho], S(\rho)) \in L^1(\mu) \times \mathbb{R}$ (`entropyEmbedding`), i.e. by the pseudometric $\max(d_1(\rho,\sigma), |S(\rho) - S(\sigma)|)$; the type synonym `EntropyDensity` carries this pseudometric. The skew Jensen–Shannon divergence $\mathrm{JS}_\lambda(\rho,\sigma) = S(\lambda\rho + (1-\lambda)\sigma) - \lambda S(\rho) - (1-\lambda)S(\sigma)$ is `skewJsd`, written as one integral.

Results.

Brief	Lean
$\int_X \lvert\rho - \sigma\rvert \, d\mu$ is the total variation of the measures with densities ρ, σ	`tvDist_withDensity_eq_integral` (ForMathlib)
eq. (2), norm reading: $\lvert p_\rho - p_\sigma\rvert(X) = d_1(\rho,\sigma)$	`tvDist_prob`
eq. (2), event reading: $\sup_A \lvert p_\rho(A) - p_\sigma(A)\rvert = \frac{1}{2} d_1(\rho,\sigma)$, attained at $A = \{\sigma < \rho\}$	`isGreatest_abs_prob_sub`
after eq. (2): L^1 convergence is total-variation convergence	`tendsto_tvDist_iff`
Lemma 4: $u^2/(2\ln 2) \le f(u) \le \lvert u\rvert$ on $[-1,1]$	`kernel_bounds`; via $f(u) = 1 - H_2(\frac{1+u}{2})$ (`kernel_eq`)
eqs. (15)–(17), pointwise	`jsdDensity_eq`, `jsdDensity_bounds`
$(\rho+\sigma)/2 \in P_\mu$ (used implicitly)	`isDensity_mid`
$\lambda\rho + (1-\lambda)\sigma \in P_\mu$ for $\lambda \in [0,1]$ (presupposed by Theorem 46)	`isDensity_mixture`; pointwise bounds `negMulLog_mixture_bounds`, via $\eta(x+y) \le \eta(x) + \eta(y)$ (`Real.negMulLog_add_le`, ForMathlib)
eq. (16): JSD as one integral	`jsd_eq_integral`
eqs. (18)–(19)	`l1dist_sq_le`: $d_1^2 \le 2\int (\rho-\sigma)^2/(\rho+\sigma)\, d\mu$
Prop. 10, eq. (11)	`jsd_bounds`
Prop. 10, eq. (12)	`tendsto_jsd_iff`, for any filter
remark after Prop. 10: JSD and total-variation convergence agree	`tendsto_jsd_iff_tvDist`
Prop. 10, last sentence: the JSD topology is the L^1 topology	`jsdTopology_eq` (balls around each point), `jsdBallTopology_eq` (generated by the balls)
the total-variation topology is the L^1 topology	`tvTopology_eq`
$\sigma \mapsto \mathrm{JSD}(\sigma,\rho)$ is L^1-continuous (needed for `jsdBallTopology_eq`)	`continuous_jsd_left`; sequentially in `tendsto_jsd_of_tendsto_l1dist`
eq. (26): $S(u_U) = \log_2 \mu(U)$	`entropy_unif`

Brief	Lean
eqs. (32)–(34): entropy of $(1-\varepsilon)u_B + \varepsilon u_A$	`entropy_mix`
eq. (31): $d_1(\rho_n, \rho) = 2\varepsilon_n$	`l1dist_mix`
Theorem 21	`entropy_not_l1_continuous`: $\rho_n \to u_B$ in L^1 and $S(\rho_n) \to S(u_B) + 1$; `not_tendsto_entropy`
second remark, eqs. (40)–(45)	`tendsto_entropy_mix_atTop`: $S(\rho_n) \to \infty$
Theorem 46	`eventually_jsd_lt` (JSD balls are τ-neighbourhoods), `eventually_l1dist_lt` (τ is finer than the L^1 topology), `eventually_tvDist_lt` (and than the total-variation topology)
Theorem 46, with the brief's hypothesis	`eventually_jsd_lt_of_mixture`, `eventually_l1dist_lt_of_mixture`, `eventually_tvDist_lt_of_mixture`, via `continuous_mid_of_continuous_mixture`
Theorem 46 for topologies (second proof)	`le_jsdBallTopology`, `le_l1Topology`, `le_tvTopology`: τ is finer; `_of_mixture` forms with the brief's hypothesis
Beyond the brief: when is S L^1-continuous? (Sec. 5)	
mixing bounds, integrated: $\lambda S(\rho) + (1-\lambda)S(\sigma) \le S(\text{mixture}) \le$ that $+ h_2(\lambda)/\ln 2$	`entropy_mixture_bounds`; `l1dist_mixture_le`: $d_1(\lambda\sigma + (1-\lambda)\rho, \rho) \le 2\lambda$
sets of arbitrarily large finite measure (*spreading*, as in Theorem 21): every ρ is an L^1 limit with $S \to +\infty$	`exists_tendsto_entropy_atTop`
sets of arbitrarily small positive measure (*concentration*): every ρ is an L^1 limit with $S \to -\infty$	`exists_tendsto_entropy_atBot`
a disjoint sequence gives spreading or concentration, and conversely	`Measure.hasDisjointSeq_iff` (ForMathlib)
Theorem 21's hypothesis gives a disjoint sequence	`Thm21Hyp.hasDisjointSeq`
with a disjoint sequence, S is L^1-discontinuous at every point	`not_continuousAt_entropy`
without one, every density is $\le M\, 1_U$ a.e., for one set U of finite measure and one M	`exists_le_indicator_of_not_hasDisjointSeq`, from `Measure.exists_measure_sdiff_eq_zero` and `Measure.exists_le_measure` (ForMathlib)
S is L^1-continuous iff there is no disjoint sequence	`continuous_entropy_iff`
S is L^1-continuous everywhere or nowhere	`continuous_entropy_or_forall_not_continuousAt`
Lebesgue measure on $[0,1]$ (not treated in the brief): S is discontinuous at every point	`not_continuousAt_entropy_Icc`, with μ the restriction of Lebesgue measure on $\mathbb{R}$ to $[0,1]$

Brief	Lean
Beyond the brief: the coarsest admissible topology (Sec. 1, Theorem 46, Sec. 5)	
$S(\lambda\rho + (1-\lambda)\sigma) =$ $\lambda S(\rho) + (1-\lambda)S(\sigma) + \mathrm{JS}_\lambda(\rho,\sigma)$	`entropy_mixture_eq`; `skewJsd_half`: $\mathrm{JS}_{1/2} = \mathrm{JSD}$
$\mathrm{JS}_\lambda(\rho,\sigma)$ is jointly continuous in λ and in ρ,σ in L^1	`tendsto_skewJsd`
mixing is continuous for the entropy topology	`continuous_mixture_entropyDensity`, `admissible_entropyTopology`
every admissible topology is finer than the entropy topology	`le_entropyTopology`
the entropy topology is the coarsest admissible topology	`isGreatest_entropyTopology`
its pseudometric vanishes exactly on densities equal a.e.	`dist_toEntropyDensity_eq_zero_iff`
it is the L^1 topology iff there is no disjoint sequence, so strictly finer under Theorem 21's hypothesis	`entropyTopology_eq_l1Topology_iff`, `entropyTopology_lt_l1Topology`

Differences from the brief.

- Densities are functions, not L^1 classes, and are required to be measurable and nonnegative everywhere rather than almost everywhere. "$S(\rho) \in \mathbb{R}$" is read as integrability of $\rho \log \rho$.
- Lemma 4 is proved through the binary entropy. With $p = (1+u)/2$ it says $(2p-1)^2/2 \le \ln 2 - H(p) \le \ln 2\,|2p-1|$. The lower bound comes from concavity of $H(p) + 2(p-\frac{1}{2})^2$, using $H'' = -1/(p(1-p)) \le -4$, which is the brief's second-derivative argument. The upper bound comes from concavity of H, the brief's chord argument.
- Eq. (18) uses Cauchy–Schwarz. The formal proof uses the pointwise inequality $|a-b| \le \frac{(a-b)^2}{(a+b)\,2c} + \frac{c(a+b)}{2}$ with $c = d_1/2$, which gives the same bound $d_1^2 \le 2\int(\rho-\sigma)^2/(\rho+\sigma)$.
- Eq. (12) is formalized through convergence along filters. The topological statements are formalized twice. First through neighbourhoods (`eventually_*`), then as equalities and inequalities of topologies on `Density`. Since JSD is not a metric, "the topology generated by the open balls" is formalized in both readings (see the setting). Theorem 46 then reads $\tau \le$ `jsdBallTopology`, the stronger of the two, which the brief's argument gives directly: continuity of $\sigma \mapsto \mathrm{JSD}(\sigma,\rho)$ makes each ball τ-open. Densities are functions, so these topologies do not separate densities that agree almost everywhere; they are not T_0.
- Eq. (2) is formalized in both readings of $\|\cdot\|_{\mathrm{TV}}$ (see the problem below). The norm reading is proved for any two integrable densities by a Hahn decomposition $\{\rho \le \sigma\}$; no σ-finiteness of μ is needed, which matters since Theorem 21 uses infinite μ.
- In Theorem 21 the hypothesis $1 < \mu(A_n)$ is dropped. Only $\mu(A_n) \to \infty$ is used, and all statements hold eventually in n. The first remark (counting measure) is not formalized separately; it is the special case X countable.
- Theorem 46 is proved in two forms. The main form assumes only that the *equal* mixture $(\rho,\sigma) \mapsto (\rho+\sigma)/2$ is continuous. That is weaker than the brief's hypothesis that all mixing

is continuous, so the formal statement is stronger. The `_of_mixture` forms take the brief's hypothesis literally, reading "statistical mixing (e.g. $(\lambda, \rho, \sigma) \mapsto \lambda\rho + (1-\lambda)\sigma$)" as this map $[0,1] \times P_\mu \times P_\mu \to P_\mu$ (`Density.mixture`), with the product topology. They follow by fixing $\lambda = \frac{1}{2}$.

- The brief asks for "the appropriate" topology on P_μ; which properties make a topology appropriate is a modelling choice and is not formalized. The mathematical part is: admissible topologies exist, there is a coarsest one, and it is Hausdorff on L^1 classes. Its pseudometric uses max; the sum $d_1 + |\Delta S|$ gives the same topology.

- The characterization of L^1-continuity is stated with disjoint sequences rather than with dimension. A disjoint sequence exists iff $L^1(\mu)$ is infinite-dimensional (the indicators 1_{A_n} are linearly independent; conversely, without a disjoint sequence the finite part of μ consists of finitely many atoms). This equivalence is not formalized.

- Literature on discontinuity. Ho and Yeung ("On the discontinuity of the Shannon information measures", IEEE Trans. Inf. Theory 55 (2009) 5362, Theorem 2) show that for every distribution P^0 of finite entropy on a countably infinite alphabet, including finitely supported ones, and every $c \in (0, \infty]$, there are P_n with $\chi^2(P^0 \,\|\, P_n) \to 0$ and $H(P_n) \to H(P^0) + c$. This is the counting-measure case of `not_continuousAt_entropy`, in a stronger mode of convergence and with any prescribed jump. Counting measure has spreading but no concentration, and discrete entropy is lower semicontinuous (their Sec. I), so only upward jumps occur there. Wehrl (Rev. Mod. Phys. 50 (1978) 221), cited there, treats the quantum analogue: von Neumann entropy is discontinuous in infinite dimension. Harremoës ("Information topologies with applications", in *Entropy, Search, Complexity*, Springer 2007, Theorem 21) shows that for discrete distributions under convergence in information divergence, which is finer than L^1, the entropy is continuous at power-dominated distributions and discontinuous at hyperbolic ones; so finer topologies do have continuity points. Piera and Parada (arXiv:0710.1275) give sufficient conditions, such as bounded densities, for convergence of differential entropy; the continuous case here is of this kind, with a uniform bound.

- Literature on the entropy topology. For discrete distributions it already appears in Ho and Yeung (Sec. VI). With $\Psi(P,Q) = H(\frac{1}{2}P + \frac{1}{2}Q) - \sqrt{H(P)H(Q)}$, $\sqrt{\Psi}$ is a metric (their Theorem 10), and $\Psi(P_n, Q) \to 0$ iff $H(P_n) \to H(Q)$ and $d_1(P_n, Q) \to 0$ (their Theorem 9). Both topologies are first countable, so on a countable alphabet the topology of $\sqrt{\Psi}$ is the entropy topology. They also use $D(Q \,\|\, P) + |H(Q) - H(P)|$. Their motivation is typicality, not mixing, and Ψ needs $H \geq 0$, so it does not extend to densities. The pseudometric $\max(d_1, |\Delta S|)$ works for every μ. We found no prior characterization of the entropy topology as the coarsest topology making mixing and entropy continuous (`isGreatest_entropyTopology`).

Problems found in the brief. No mathematical errors. One notational ambiguity and one understatement:

- Eq. (2), $\|p_\rho - p_\sigma\|_{\mathrm{TV}} = \frac{1}{2}\|\rho - \sigma\|_{L^1(\mu)}$, does not say which total variation is meant. With the usual total variation norm $|p_\rho - p_\sigma|(X)$, which is Mathlib's `tvDist`, the factor $\frac{1}{2}$ is wrong: $|p_\rho - p_\sigma|(X) = \|\rho - \sigma\|_{L^1(\mu)}$ (`tvDist_prob`). The factor is right for the distance $\sup_A |p_\rho(A) - p_\sigma(A)|$ (`isGreatest_abs_prob_sub`). Nothing downstream depends on the constant, since only convergence is used.

- The conclusion extends the failure of L^1-continuity to "typical infinite-dimensional spaces". In fact it fails at every point of every P_μ for which μ has a disjoint sequence, i.e. $L^1(\mu)$ is infinite-dimensional (`not_continuousAt_entropy`), and holds otherwise (`continuous_entropy_iff`). Theorem 21 covers only spreading. Lebesgue measure on $[0,1]$, where every set has finite measure, is a concentration case: mixing in a little of the uniform density on a short interval sends S to $-\infty$.

Gaps the formalization had to fill:

- Prop. 10 and Theorem 46 use $(\rho+\sigma)/2 \in P_\mu$, i.e. that the mixture has finite entropy. The brief does not prove this. It follows from the pointwise bounds $0 \le$ JSD-integrand $\le \frac{\ln 2}{2}|\rho-\sigma|$ (`isDensity_mid`).

- Theorem 46 presupposes that P_μ is closed under mixing, which the brief does not prove either. For $m = \lambda a + (1-\lambda)b$ and $\eta(x) = -x\ln x$, concavity and subadditivity of η give $0 \le \eta(m) - \lambda\eta(a) - (1-\lambda)\eta(b) \le \eta(\lambda)\,a + \eta(1-\lambda)\,b$, so $\eta(m)$ is integrable (`isDensity_mixture`).

- The last sentence of Prop. 10 needs more than eq. (11) in the second reading. The bounds make each JSD ball an L^1 neighbourhood of its centre, which gives `jsdTopology_eq`. For the topology generated by the balls, each ball must be L^1-open, i.e. $\sigma \mapsto \mathrm{JSD}(\sigma,\rho)$ must be L^1-upper semicontinuous; the brief does not address this. The formal proof shows full continuity. For $\sigma_n \to \sigma$ in L^1, a subsequence converges a.e. (via convergence in measure). The integrand g_n is split into $\min(g_n, G)$, with $G = \frac{\ln 2}{2}(\sigma+\rho)$, which converges by dominated convergence, and the excess $g_n - \min(g_n, G)$, which is at most $\frac{\ln 2}{2}|\sigma_n - \sigma|$ by eq. (17). This is the generalized dominated convergence theorem, with dominating functions $\frac{\ln 2}{2}(\sigma_n + \rho)$ converging in L^1 (`MeasureTheory.tendsto_integral_of_dominated_of_tendsto_L1`, ForMathlib; `tendsto_integral_jsdDensity`).

- In Theorem 21, $\varepsilon_n = 1/\log_2 \mu(A_n) \in (0,1)$ needs $\mu(A_n) > 2$, not $\mu(A_n) > 1$. The brief's "passing to a tail" covers this.